BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Functions and Return Values

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What is a record?

2. Why retest a program after a fix?

A) Check for new faults
B) Prove no faults can exist
C) Avoid testing old features
D) Remove all input rules

3. "The Internet is a single computer containing every website." Correct this. Explain why.

4. A browser asks for a school's web page. "A client must host a website before asking for it." Correct this. Which computer provides the page?

Today's objectives

Define functions with parameters and return values. Use the technique in a different problem.

Key vocabulary - Read, cover, recall

Function
Named reusable subprogram that returns a value to its caller.

Parameters
Named inputs that allow a function to work on supplied arguments.

Return
A RETURN statement sends a calculated value to the calling code.

Modularity
Dividing a program into functions makes sections easier to reuse and test.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Function	Named reusable subprogram that returns a value to its caller. In the rectangle-area calculator, identify where this technique applies.
Parameters	Named inputs that allow a function to work on supplied arguments. In the taxi fare function, identify where this technique applies.
Return	A RETURN statement sends a calculated value to the calling code. In the test-mark average, identify where this technique applies.
Modularity	Dividing a program into functions makes sections easier to reuse and test. In the triangle area, identify where this technique applies.

I Do - One worked example

A rectangle-area calculator uses function and parameters. Explain what each contributes. [4 marks]

1. Identify the program's purpose and the pieces of data needed.

2. Function [1]; Named reusable subprogram that returns a value to its caller. [1].

3. Parameters [1]; Named inputs that allow a function to work on supplied arguments. [1].

4. Each feature needs a correct technical fact plus an explanation.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A taxi fare function needs function. Explain the purpose. [2 marks]

2. Explain parameters for a test-mark average. [2 marks]

3. When programming a triangle area, explain why return and modularity matter. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the return statement so the function calculates a pixel count.

```
function pixels(width, height)
return ...
endfunction
print(pixels(4, 3))
```

BLACK PEN: predict or complete the code; test, then improve.

You Do - Independent exam questions

Q2. State TWO relevant ideas or operations from this topic. [2 marks]

Q3. Explain parameters and return in a shopping cost function. [4 marks]

Q4. A programmer makes a class total calculator. Explain the roles of function, return and modularity. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A function is a named reusable subprogram that returns a value to its caller. ☐ T ☐ F
2. Named reusable subprogram that returns a value to its caller ☐ T ☐ F
3. Named inputs that allow a function to work on supplied arguments ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Named fields about one item or person. Fields may have different data types. ☐

Q2 A - Check for new faults ☐

Q3 A worldwide collection of interconnected computer networks. ☐

Q4 A web server stores website resources and sends them when clients request them. Access needs working servers and network services. ☐

Code First - Model

```
function pixels(width, height)
return width * height
endfunction
print(pixels(4, 3))
```

Output: 12

We Do - Check the model

1. Function [1]; Named reusable subprogram that returns a value to its caller. [1]. ☐

2. Parameters [1]; Named inputs that allow a function to work on supplied arguments. [1]. ☐

3. Return [1]; A RETURN statement sends a calculated value to the calling code. [1]. Modularity [1]; Dividing a program into functions makes sections easier to reuse and test. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Function [1]; Parameters [1]. ☐

Q3 Parameters [1]; Named inputs that allow a function to work on supplied arguments. [1]. Return [1]; A RETURN statement sends a calculated value to the calling code. [1]. ☐

Q4 Function [1]; Named reusable subprogram that returns a value to its caller. [1]. Return [1]; A RETURN statement sends a calculated value to the calling code. [1]. Modularity [1]; Dividing a program into functions makes sections easier to reuse and test. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: